

Differentiable functions – 7

Extrema and optimisation

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Theoretical minimum.

Theorem 1 (Fermat and the second-order test). At an interior extremum of a differentiable function the derivative vanishes; if $f'(x_0) = 0$ and $f''(x_0) > 0$, the point is a strict local minimum. In several variables the gradient vanishes and the Hessian is positive semidefinite.

Theorem 2 (Lagrange multipliers). If f has a local extremum on $\{g = 0\}$ at a point where $\nabla g \neq 0$, then $\nabla f = \lambda \nabla g$ for some λ . With several constraints the gradient of f lies in the span of the gradients of the constraints.

Fact 1. Before differentiating, secure existence: on a compact set continuity is enough, on a non-compact one use coercivity or a symmetry argument. Half of the wrong solutions to optimisation problems are wrong exactly here.

1. Prove Fermat's theorem and the second-order test, and give an example of a smooth function with $f'(x_0) = f''(x_0) = 0$ which has a minimum, a maximum and no extremum at x_0 in three different cases.
2. Prove the Lagrange multiplier rule in the case of one constraint, and use it to find the extrema of $x_1 x_2 \cdots x_n$ on the sphere $x_1^2 + \cdots + x_n^2 = 1$, deducing the AM–GM inequality.
3. Among all triangles inscribed in a given circle find the one of maximal area, and among all triangles with a given perimeter the one of maximal area; in both cases prove the existence of the maximum before computing it.
4. Find $\min_{a,b \in \mathbb{R}} \int_0^1 |x^2 - ax - b|^2 dx$ and $\min_{a,b} \max_{x \in [0,1]} |x^2 - ax - b|$, and explain why the answers to the two problems are attained at different pairs (a, b) .

5. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous with $f(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$. Prove that f attains its minimum, and apply this to show that every polynomial in two variables which is positive and tends to infinity along every ray attains a global minimum.